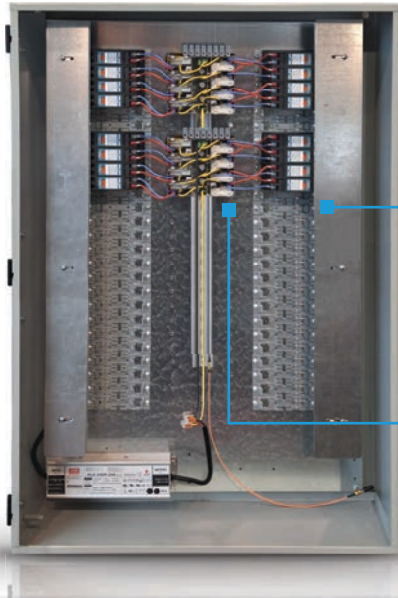


Relay Control Panel Assembly

48 Relay Capacity

Easily and Quickly Replace Lighting Control Panels



Full Range of Control Features

Time clock, motion sensing, wall stations, daylight harvesting, BACnet interfaces, remote commissioning and operation, floorplan view control, and more

Fully Assembled Panels

Assemblies come pre-wired with 8, 16, 24, 32, 40 or 48 relays, in UL-508 listed configurations.

PRODUCT OVERVIEW

Description

Each unit comes pre-wired with 8, 16, 24, 32, 40 or 48 relays, fully assembled in UL-listed configurations. The panels are delivered fully assembled, including all control components, making installation easy.

Control functions include relay control, 0–10V dimming, time clocks, scenes, daylight harvesting, motion sensing, and a wide array of AC, low-voltage, and battery-powered wall stations—**everything needed to comply with the latest energy codes.**

Motion sensing, wall stations, and other distributed components communicate over Avi-on's wireless mesh, **eliminating the need to run any control system wiring outside the panel** (other than wired 0–10V dimming).

Avi-on's controls meet the highest standards of security and performance—both wired and wireless—with UL's highest security rating (UL 1376 Diamond).

Programming and support can be handled either locally or remotely using a mobile device or laptop, just like any other Avi-on system.

Features

Emergency and non-emergency circuits can be placed in the same panel (with an optional divider) using Avi-on's unique UL 924 beacon solution. Every individual relay controller is UL 924 rated and can be designated in software as an emergency circuit.

You can also mix voltages in the panel as needed (with an optional divider).

Easily expand by purchasing individual relays and control units (control units are packaged in blocks of 8).

Use Avi-on's Ethernet-to-Mesh Link to bridge panel rooms over existing (or new) Ethernet wiring, or bridge them using Avi-on's wireless mesh—your choice.

Benefits

With fast, straightforward installation and no control system wiring (other than 0–10V dimming), Avi-on offers the perfect balance of familiar relay panel-based lighting controls while still providing the benefits of a fully distributed wireless control system. It is upgradable and expandable to full enterprise-level capabilities—**all without the need for new wiring or gateways, ever.**

COMPATIBILITY

Should replace any system with equal or fewer relays. Contact Avi-on if you are replacing a panel from another manufacturer.

Project		Location/Type	
---------	--	---------------	--

ORDERING INFORMATION

Part Number	Description	Application	Input Voltage	Weight*
 AVI-PNL-360-8-277	Relay Panel Interface Assembly - 8 Circuit block 110-277 VAC	Indoor	110-277 VAC	85 lbs
 AVI-PNL-360-18-277	Relay Panel Interface Assembly - 16 Circuit block 110-277 VAC	Indoor	110-277 VAC	90 lbs
 AVI-PNL-360-24-277	Relay Panel Interface Assembly - 24 Circuit block 110-277 VAC	Indoor	110-277 VAC	95 lbs
 AVI-PNL-360-32-277	Relay Panel Interface Assembly - 32 Circuit block 110-277 VAC	Indoor	110-277 VAC	100 lbs
 AVI-PNL-360-40-277	Relay Panel Interface Assembly - 40 Circuit block 110-277 VAC	Indoor	110-277 VAC	105 lbs
 AVI-PNL-360-48-277	Relay Panel Interface Assembly - 48 Circuit block 110-277 VAC	Indoor	110-277 VAC	110 lbs

*shipping weight

To order please contact Avi-on sales at **(877) AVION-US**, (877) 284-6687 or prosales@avi-on.com for information on becoming an Avi-on partner and order details.

CONTROL SPECIFICATIONS

RELAY CONTROL ASSEMBLY

Input Voltage

(Power supplies): 90-305VAC (277 VAC version)

1 Power supply per two 8 relay assemblies (included)

Physical Parameters:

Mounting:

DIN Rail Clip (included)

Antenna:

Single Remote Antenna
Frame Mount RPSMA cable
with included Magnetic
Antenna (coming soon)

0-10V Dimming:

100mA per Channel

Operating Temperature:

14F to 122F (-10C to +50C)

Storage Temperature:

-40F to +185F (-40C to +85C)

Humidity Rating:

95% non-condensing

Radio Frequency:

2.4GHz

Wireless Standard:

BLE 5.0 with Mesh

Point to Point Range*:

25m (80ft) with obstructions and
107m (350ft) unobstructed

Security:

AES 128-bit encryption for device to device
communication
AES 256-bit encryption for device to cloud
communication

Warranty:

5 years; 10 years optional

Regulatory:

UL: UL924-2022 (ED10)
UL 60730-1 (ED5)

FCC: FCC: 2AFZI-AVIBG21
FCC Part 15, Subpart B (Class B)
FCC Part 15.247

IC: IC: 20544-AVIBG21
ICES-003, Issue 7, Oct. 2020
RSS-GEN Issue 5, Feb. 2021 Amendment 2
RSS-247 Issue 2, Feb. 2017

BQB: DID: D063032
Qualified Design ID (QDID):
205509
178212
175341

**When communicating through the mesh, range is essentially unlimited (5000ft+)
Repeaters may be necessary if the nearest switches or sensors are located more than
25m (80 feet) from the panel*

PANEL SPECIFICATIONS

NEMA/Type:

1

Mounting:

Surface Mount

Weight:

84 lbs. 4 oz

Dimensions:

36.00" H x 24.00" W x 6.00" D

Capacity:

52 Relays

Subplate:

Laser Cut

Cover Type:

Hinged Locking Cover

INCLUDED SUBCOMPONENTS

ENC-361

NEMA/Type 1 Enclosure

SP-361

Subplate (34"H x 22"W x .5"D)

FB-36

Finger Barriers

BR-SO

Stand-off Brackets

HW-KDL

Key Locking Door Latch

CRC9000

Mounting Straps

Accessories

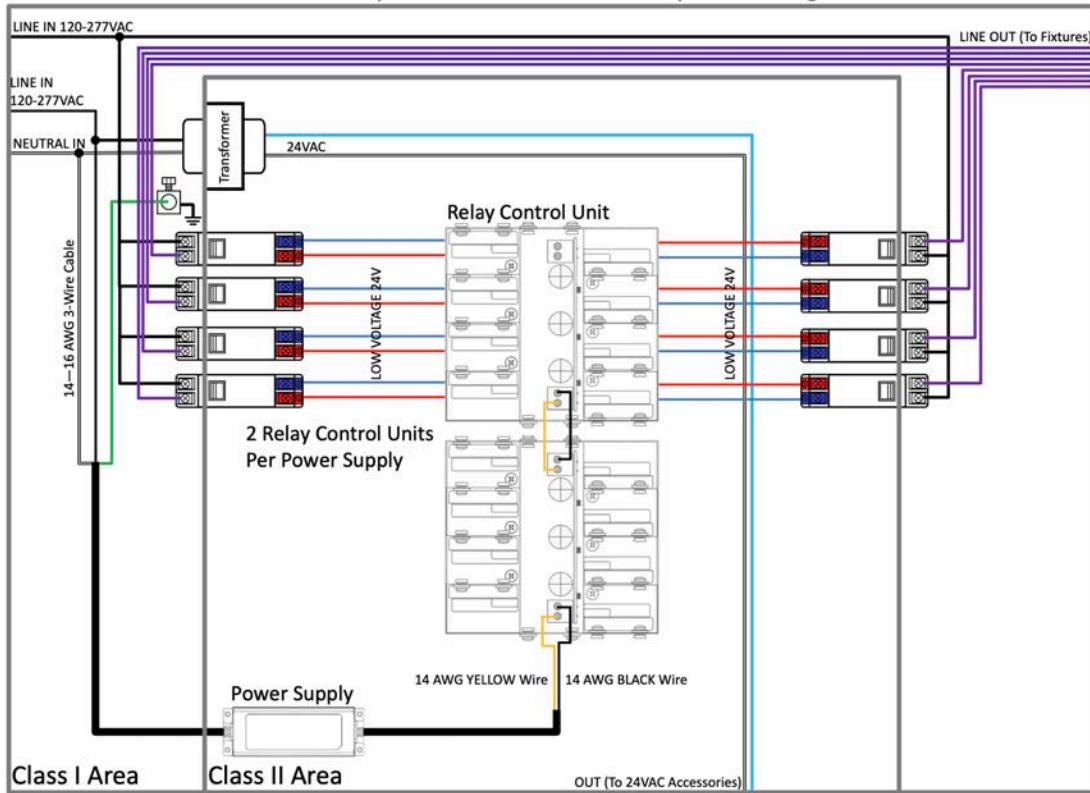
VB-1

Voltage Barrier (optional)

TB-2-6-8

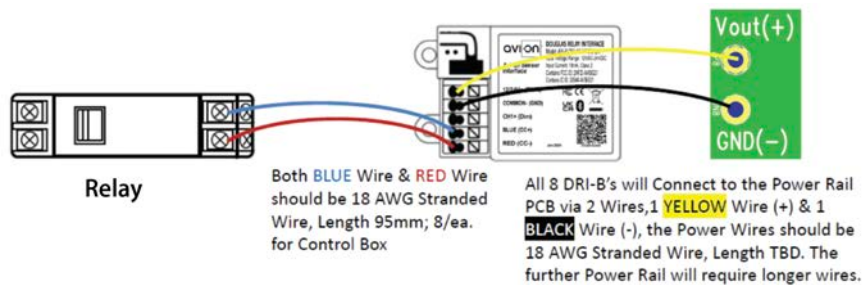
Terminal Block Assembly
(8 Line Voltage, 6 Low Voltage)

WIRING DIAGRAMS



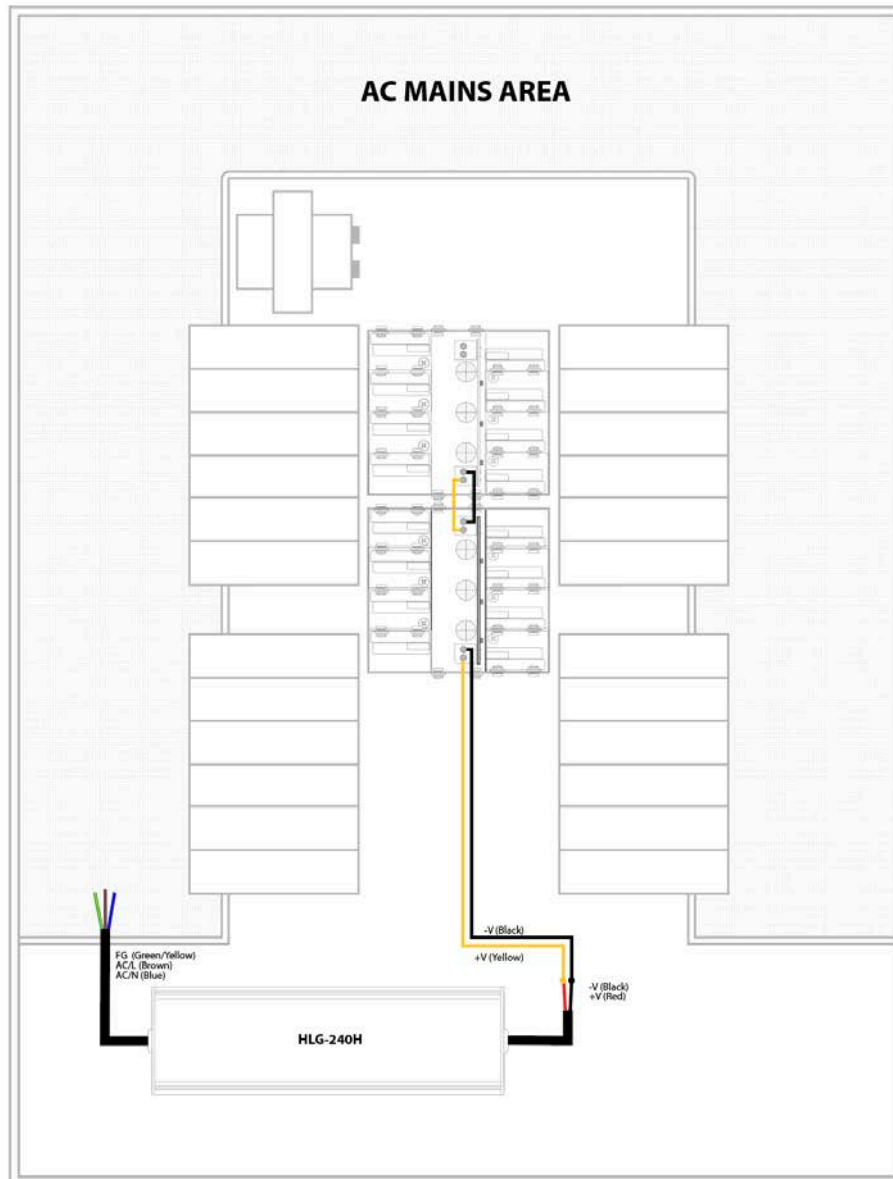
Avi-on Relay Control Panel Assembly Riser Diagram
(2 Relay Control Units Per Power Supply)

RCI-B & RELAY WIRING



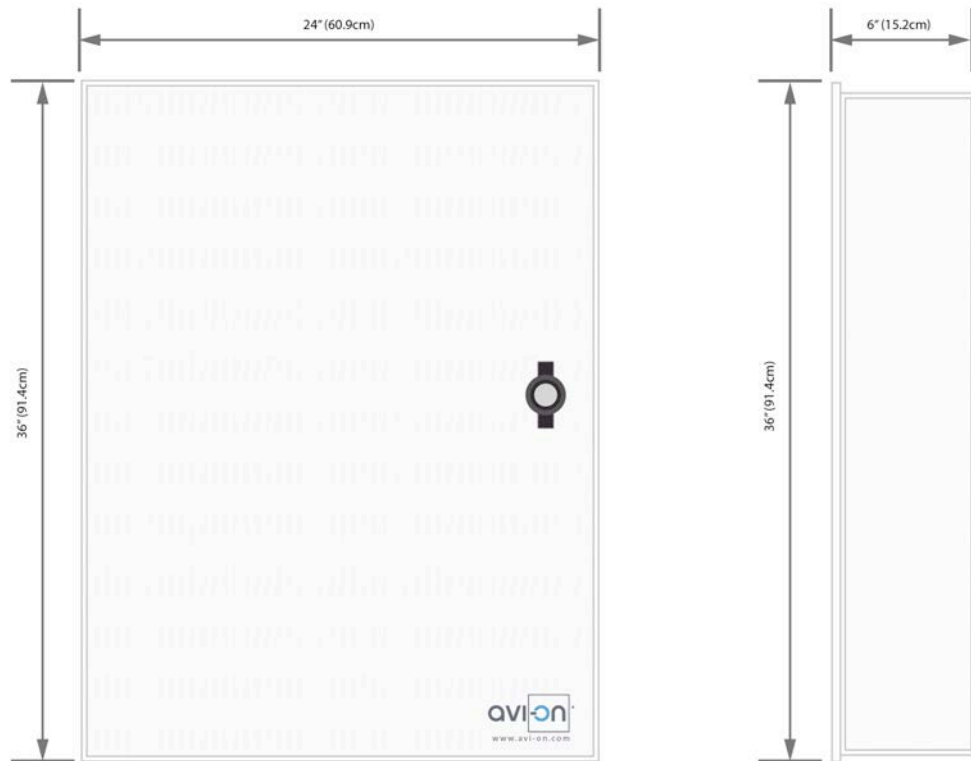
Internal Assembly Wiring Diagram

WIRING DIAGRAMS



Assembly Wiring Diagram

DIMENSIONS



Assembly Dimensions

ALL PRODUCT, PRODUCT SPECIFICATIONS AND DATA ARE SUBJECT TO CHANGE WITHOUT NOTICE TO IMPROVE RELIABILITY, FUNCTION OR DESIGN OR OTHERWISE. The information contained herein is believed to be reliable. Avion makes no warranty, representation or guarantee regarding the information contained herein, the suitability of the products for any particular purpose, or the continuing production of any product. Avion assumes no responsibility or liability whatsoever for the use of the information contained herein. The information contained herein, or any use of such information does not grant, explicitly or implicitly, to any party any patent rights, licenses, or any other intellectual property rights, whether with regard to such information itself or anything described by such information.

